

ALU de 2 bits

S_1 (Sel 1)	S_0 (Sel 0)	A (Op 1)	B (Op 2)	R (Resultado)	C (Acarreo)	Operación Ejecutada
0	0	0	0	0	0	AND (0 · 0)
0	0	0	1	0	0	AND (0 · 1)
0	0	1	0	0	0	AND (1 · 0)
0	0	1	1	1	0	AND (1 · 1)
0	1	0	0	0	0	OR (0 + 0)
0	1	0	1	1	0	OR (0 + 1)
0	1	1	0	1	0	OR (1 + 0)
0	1	1	1	1	0	OR (1 + 1)
1	0	0	0	0	0	XOR (0 ⊕ 0)
1	0	0	1	1	0	XOR (0 ⊕ 1)
1	0	1	0	1	0	XOR (1 ⊕ 0)
1	0	1	1	0	0	XOR (1 ⊕ 1)
1	1	0	0	0	0	SUMA (0 + 0)
1	1	0	1	1	0	SUMA (0 + 1)
1	1	1	0	1	0	SUMA (1 + 0)
1	1	1	1	0	1	SUMA (1 + 1 = 2, R=0, C=1)

Expresion booleana y simplificacion
$R = \overline{S_1} \cdot \overline{S_0} \cdot (A \cdot B) + \overline{S_1} \cdot S_0 \cdot (A + B) + S_1 \cdot \overline{S_0} \cdot (A \oplus B) + S_1 \cdot S_0 \cdot (A \oplus B)$
$R = \overline{S_1}(\overline{S_0}(A \cdot B) + S_0(A + B)) + S_1 \cdot (A \oplus B) \cdot (\overline{S_0} + S_0)$
$R = \overline{S_1} \cdot \overline{S_0} \cdot (A \cdot B) + \overline{S_1} \cdot S_0 \cdot (A + B) + S_1 \cdot (A \oplus B)$
$C = S_1 \cdot S_0 \cdot (A \cdot B)$